Computer Science & IT

SIRX

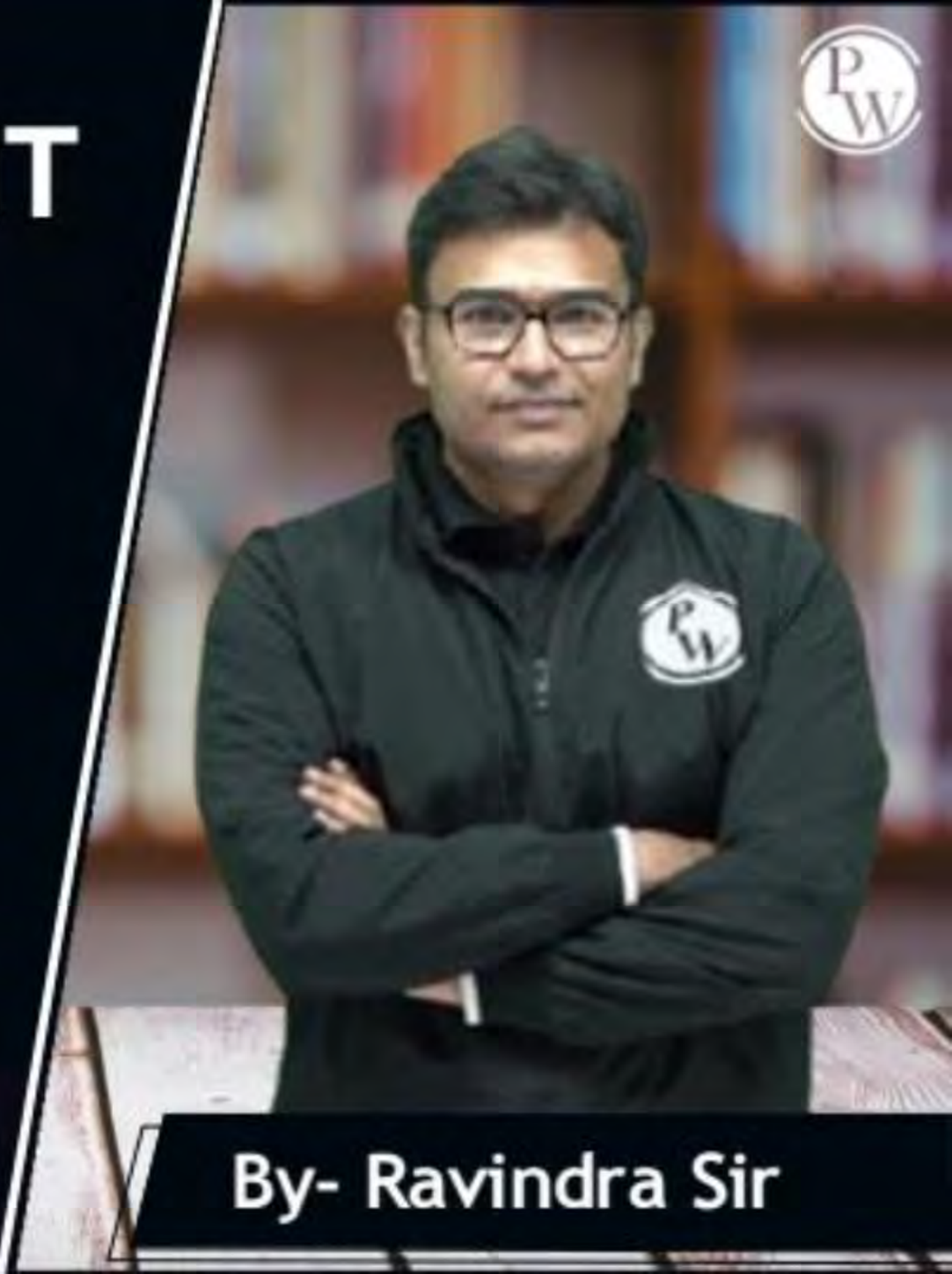
Anna

COMPUTER NETWORKS

(CN)

Flow Control Methods

Lecture No. 03



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

SWP

Topics to be Covered



- Topic
- Topic
- Topic

SWP



Inspiring Stories : Percy Julian

Percy Julian – The Chemist Who Made Steroids Affordable

Background: An African-American scientist born into segregation-era Alabama.

Struggles: Faced deep racial bias in education and employment, repeatedly denied recognition and opportunity.

Achievements: Pioneered the synthesis of cortisone and other steroids—transforming medicine globally. Held over 130 patents.

Impact: He synthesized breakthroughs even while discriminated against and silenced. If he could invent in the face of exclusion, you can code through chaos with clarity.



Inspiring Stories : Geetha Manjunath

Geetha Manjunath Background: Grew up in Bengaluru without any notable startup ecosystem exposure. Her journey into AI and healthcare began with sheer curiosity and determination.

Education: Earned her Master's in Computer Science from IISc Bangalore and later completed a Ph.D. there in AI and Data Mining. She also holds an executive degree from Kellogg School of Management.

Career Achievements: Founded NIRAMAI Health Analytix, a Bengaluru-based health-tech startup that developed a radiation-free AI method for early breast cancer detection.

Impact: Revolutionizing cancer screening with accessible, privacy-preserving technology, Geetha has made diagnostics safer, more scalable, and more inclusive, especially vital for women in low-resource settings.



Inspiring Stories : Malvika Iyer

Malvika Iyer – The Scholar Who Refused to Be Limited

Background: A young girl from Tamil Nadu, India. Lost both her hands in a grenade accident at age 13.

Struggles: Faced immense physical and social obstacles, disability, stigma, and doubt at every corner.

Achievements: Earned a Ph.D. in Social Work, became a global disability rights advocate, and inspires with accessible fashion and inclusivity.

Impact: She didn't just survive her trauma; she studied it.

If she could earn a doctorate without hands, you can nail GATE with just head, heart, and hard work.



Inspiring Stories : Pavan Guntupalli

Pavan Guntupalli (IIT Alumnus)

Background: Faced over 75 investor rejections early in his entrepreneurial journey.

Education: Graduate of an IIT (specific branch not confirmed in sources).

Career Achievements: Co-founded Rapido, a ride-hailing platform that soared to a ₹9,500 crore business within a decade.

Impact: Became a leading force in India's mobility sector, transforming urban transport—his grit reshaped an industry despite repeated early setbacks.

Stop and wait

+

ARQ

+

Seq num → Data

+

Seq num → ACK

$\frac{1}{1+2a}$ ⇒ pipelining

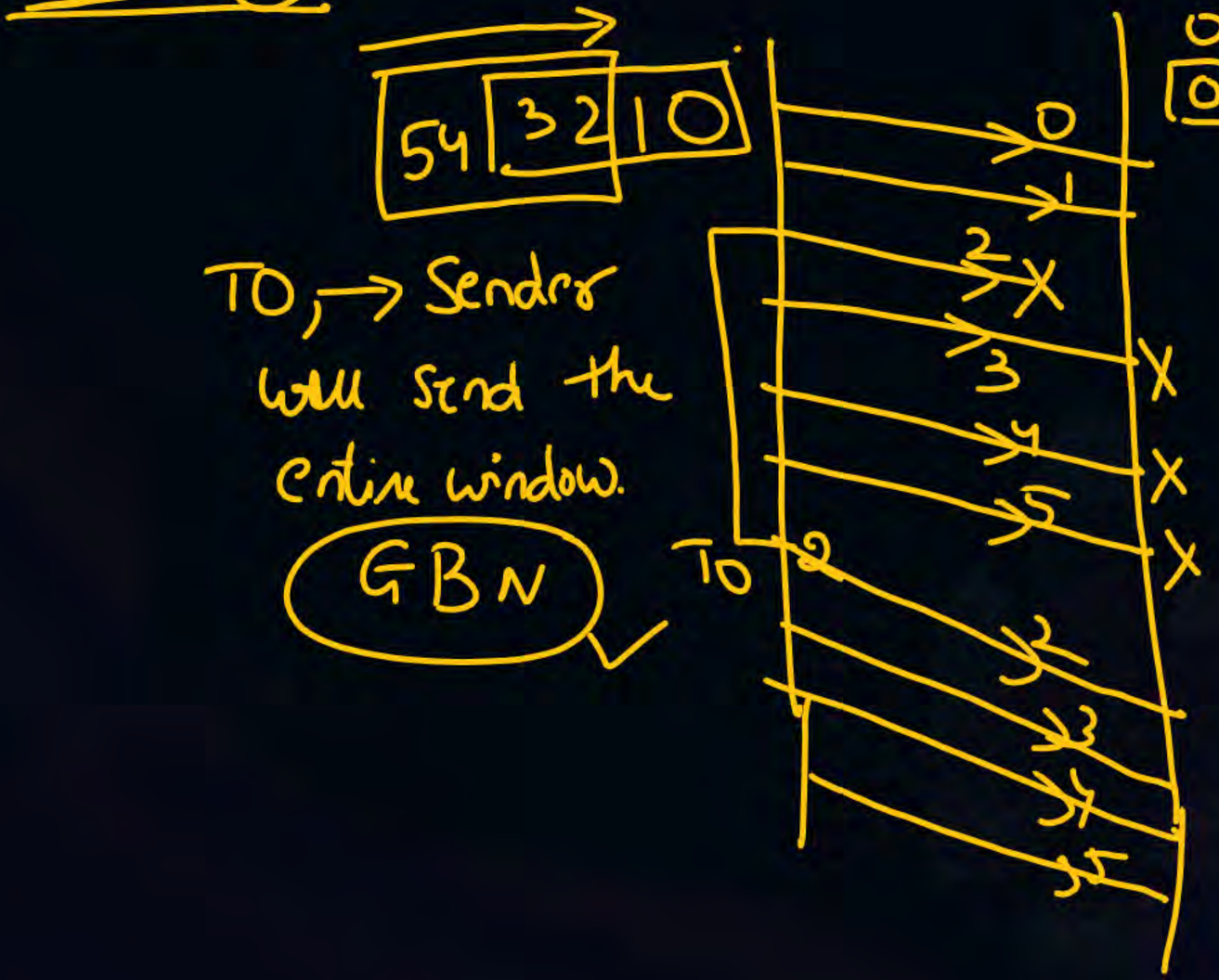
SWP

GBN

SR

RBR-1
RBR-2
RBR-3
RBR-4

GBN ✓



$TO_1 \rightarrow$ Sender will send the entire window.

GBN

GBN:-

Rule 3: Acknowlegments

Idependent



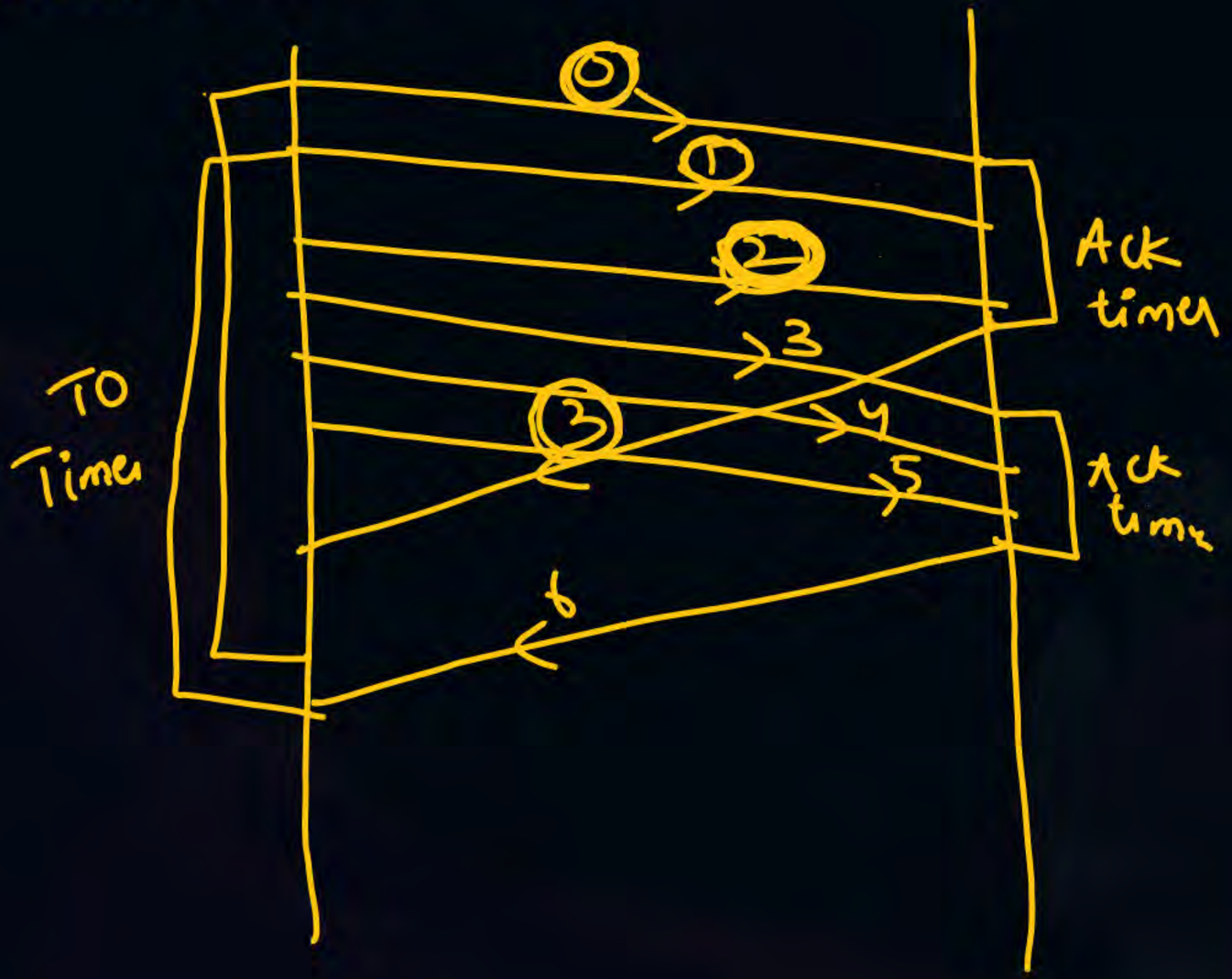
Adv: Reliable

Disadv: Traffic ↑

Cummulative



GBN:- uses cummulative ack



RBR P1: $W_S = N$

P2: $W_R = 1$

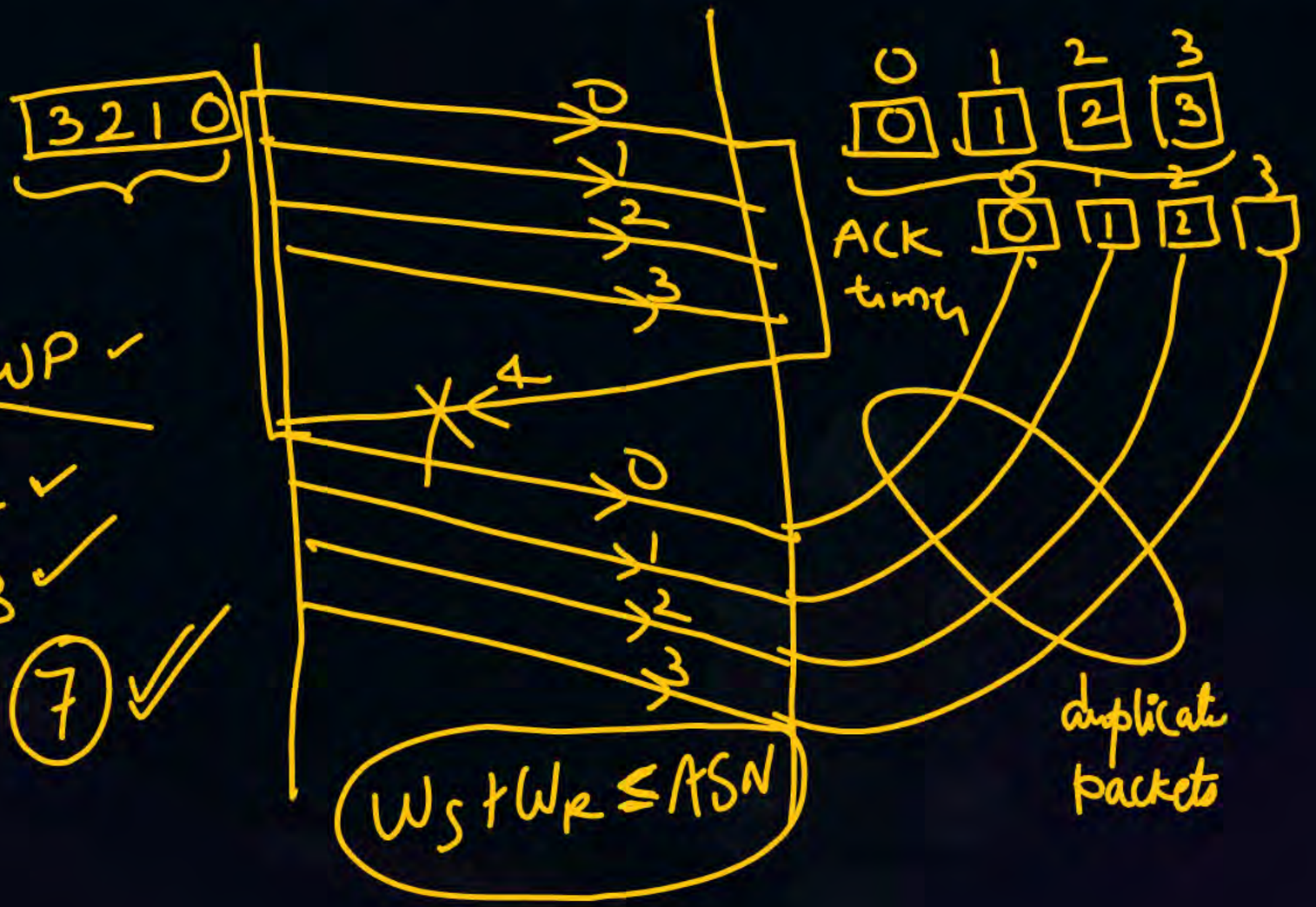
P3: Ack - cumm

P4: Seq numb

Seq number

GB4

Seq num = 4

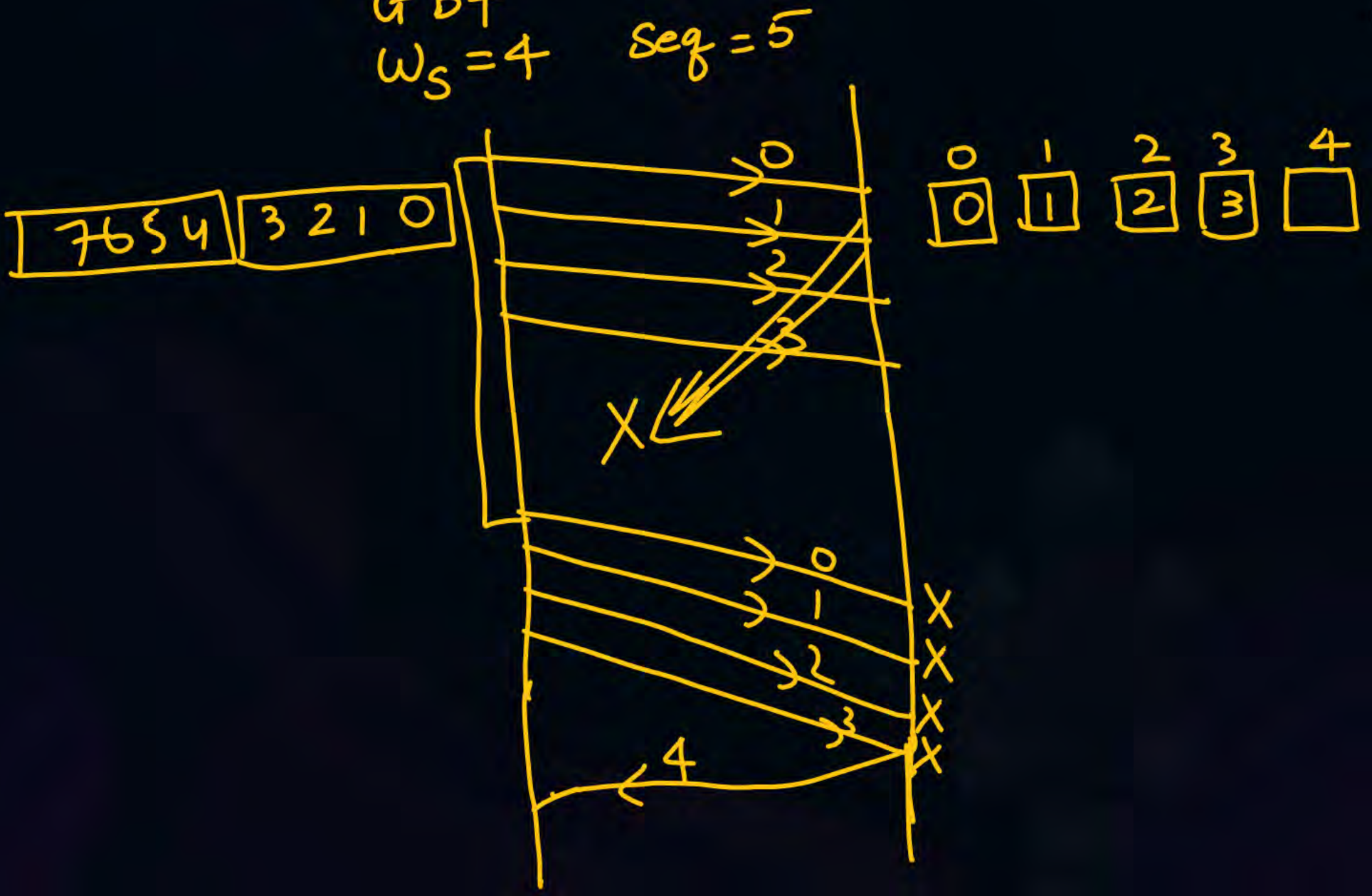


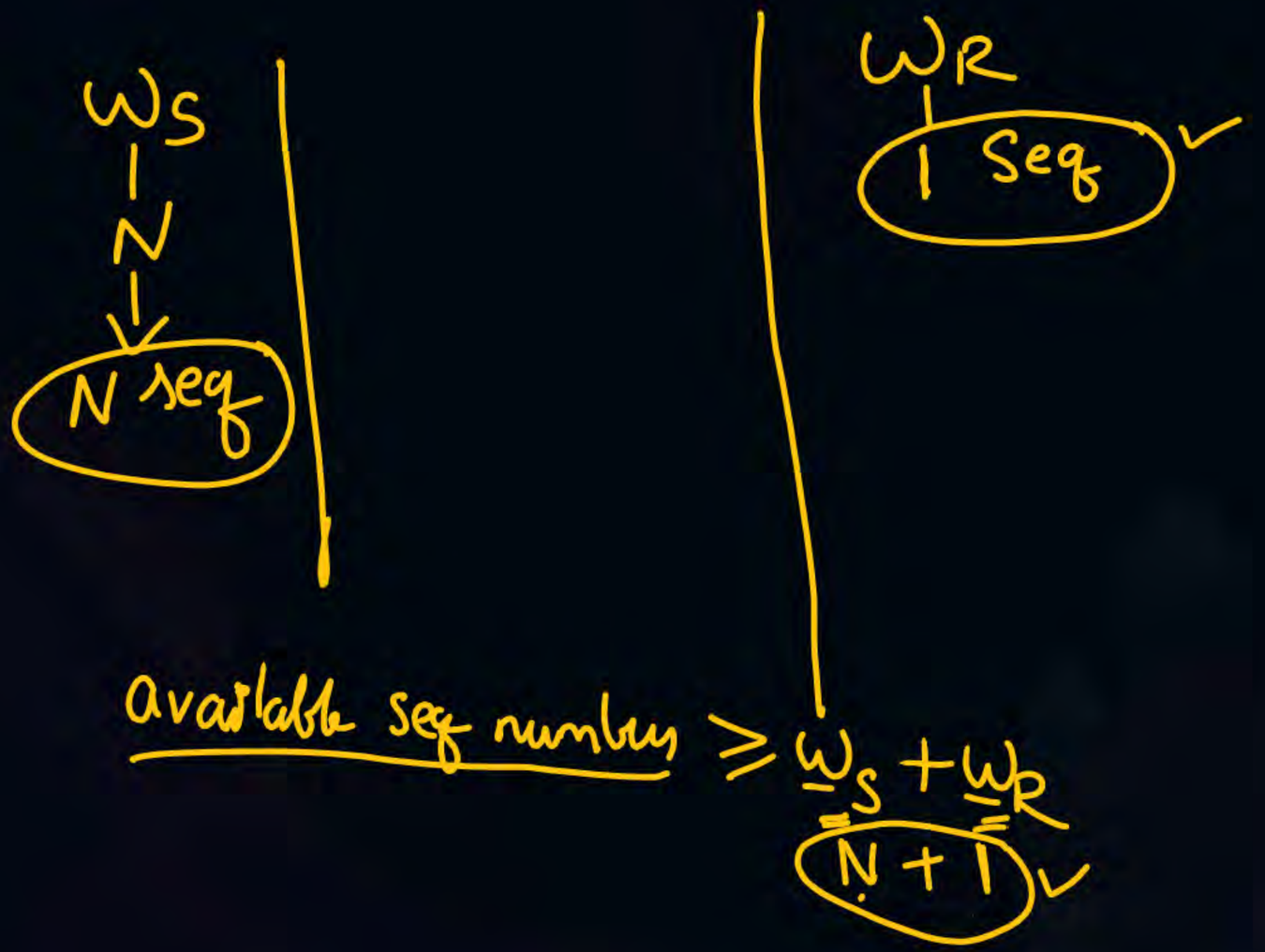
Mirage SWP ✓

$W_S = 4$ ✓

$W_R = 3$ ✓

min Seq = ? (7) ✓✓


GBN
$W_S = 4$
Seq = 5
7654
3210
0
1
2
3
X
0
1
2
3
4
X
X
X
X
0
1
2
3
4


Ws
N
N seq
WR
1 seq
available seq number ≥ Ws + WR
N + 1

GBN

$W_S = N \quad W_R = 1$

min Seq numbs $= (N+1)$

Bits in Seq numbs field $= \lceil \log_2(N+1) \rceil$

$Seq \geq W_S + W_R$

$2 \quad | \quad 1$



If Seq numbers available are 'N'

then $w_S = ?$ $w_R = ?$ GBN

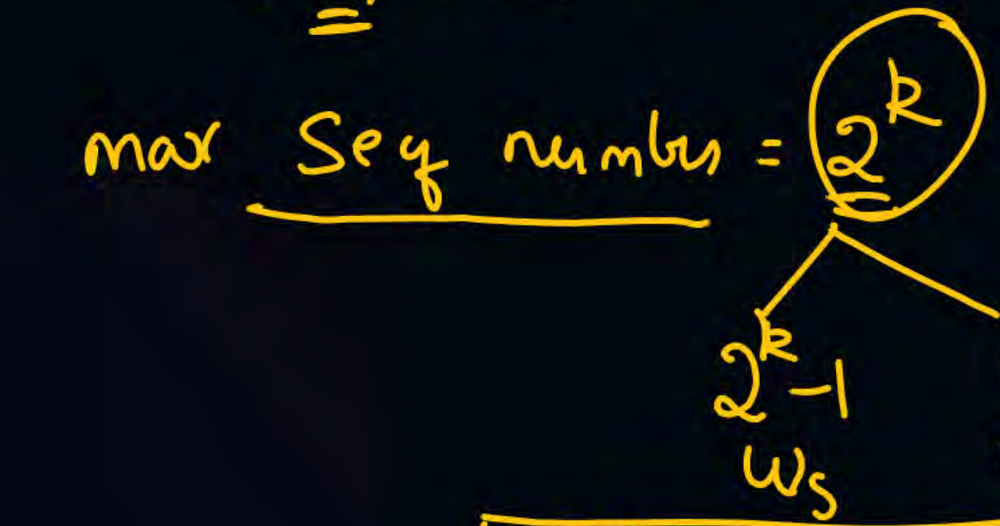
$w_S = N-1$ $w_R = 1$

$w_{S\ max} = N-1$

If the no of bits in Seq number field = k, then what is max W_S and max W_R in GBN?

max Seq number = 2^k

$W_S + W_R \leq 2$



$W_S + W_R \leq$ available sequence number

SR

RBR R-1

RBR R-2

RBR R-3

RBR R-4

S R
/ \
Selective Repeat :=

RBR 1) $W_S = N$, $N > 1$

BW = 4 mbps

Th = ? $\eta * BW$

$\frac{1}{2} * 4\,mbps = 2\,mbps$

$T_t = 1\,ms$, $T_p = 49.5\,ms$, what is the efficiency of SR with

$W_S = 50$

$1 + 2a = 1 + 2 * T_p/T_t = 1 + 2 * \frac{49.5\,ms}{1\,ms}$

$= 100$

$\eta = \frac{|W_S|}{1+2a} = \frac{50}{1+2a} = \frac{50}{100} = 50\% = \frac{1}{2}$

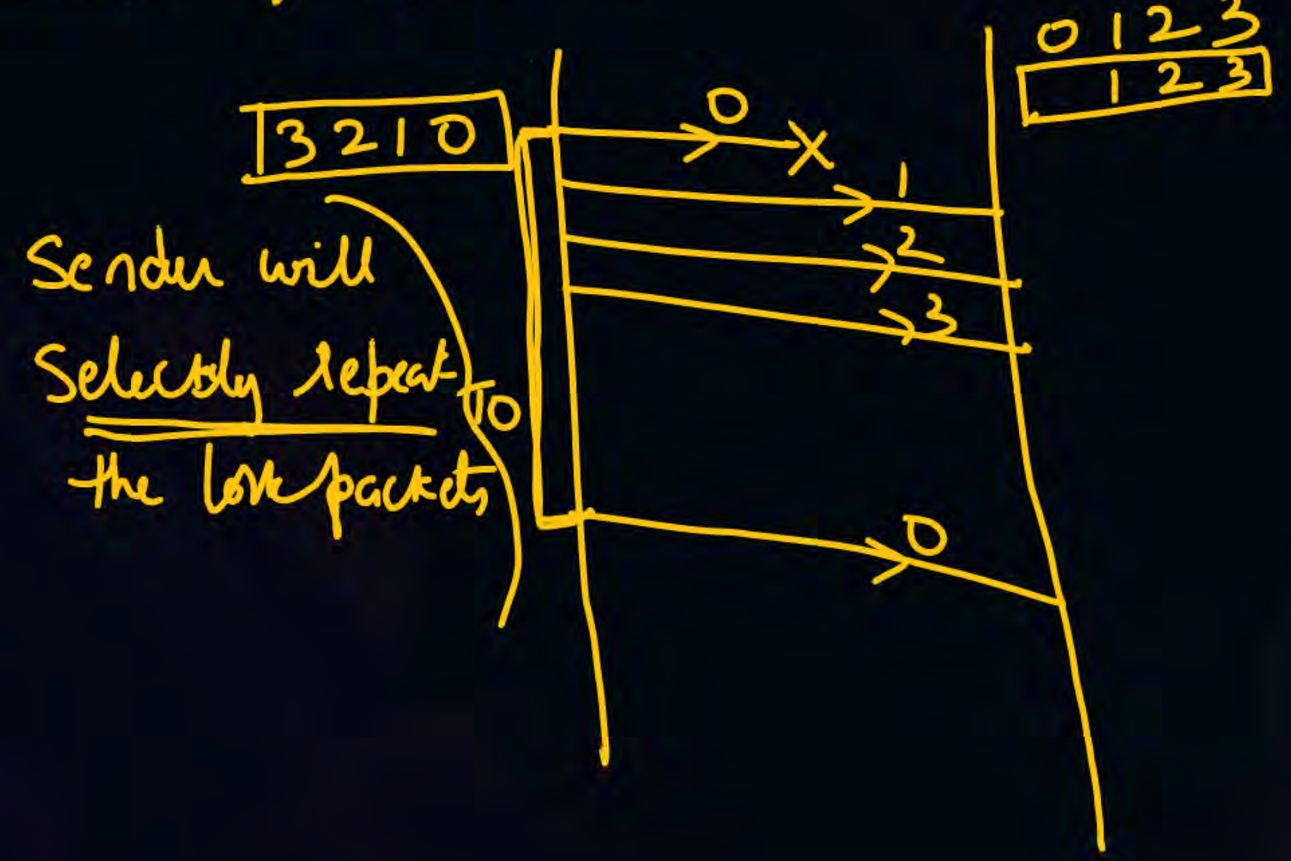
In SR, $(W_S = W_R = 3)$ 10 packets are transmitted every 5th packet is lost, then how many total transmissions

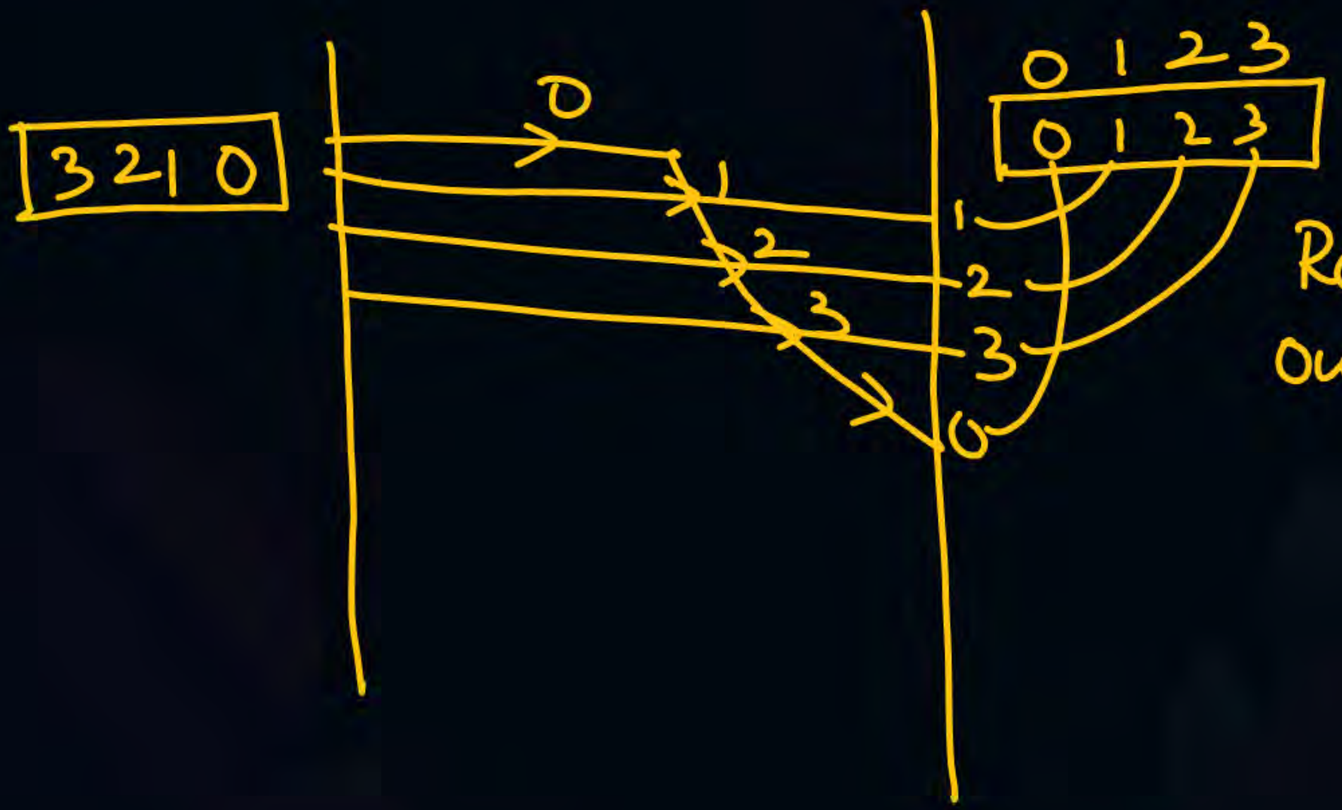
1 2 3 4 5 5 6 7 8 9 9 10

(12)

SR

Rule 2 :- $|W_R| = |W_S|$





Rec Can Rec
Out of order packets

SR

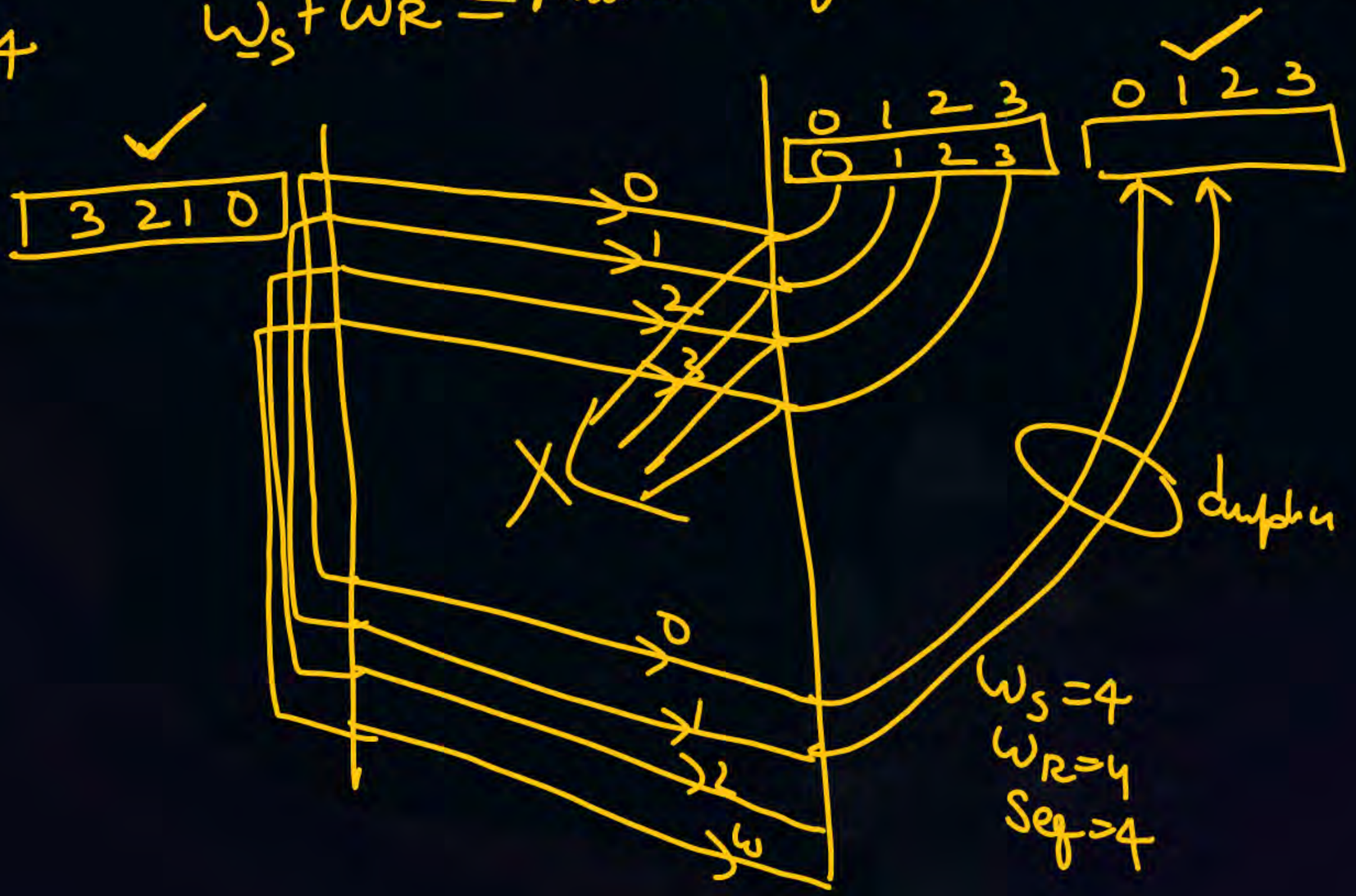
Rule 3:- SR uses independent ack



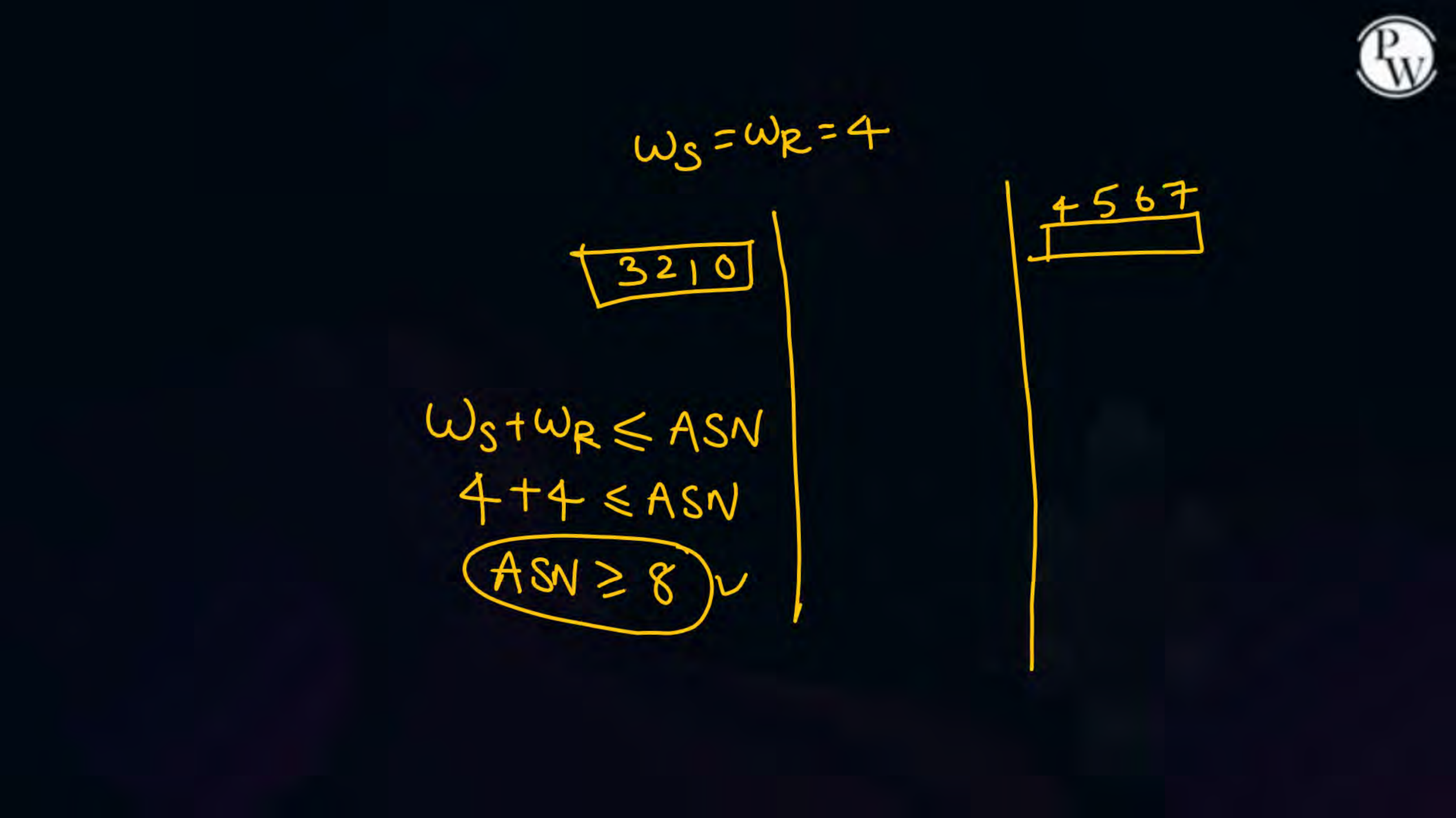

0
1
X
2
3
4
0 1 2 3

SR
RBR - R4

$W_S + W_R \leq$ Available seq number



$W_S = 4$
$W_R = 4$
Seq = 4

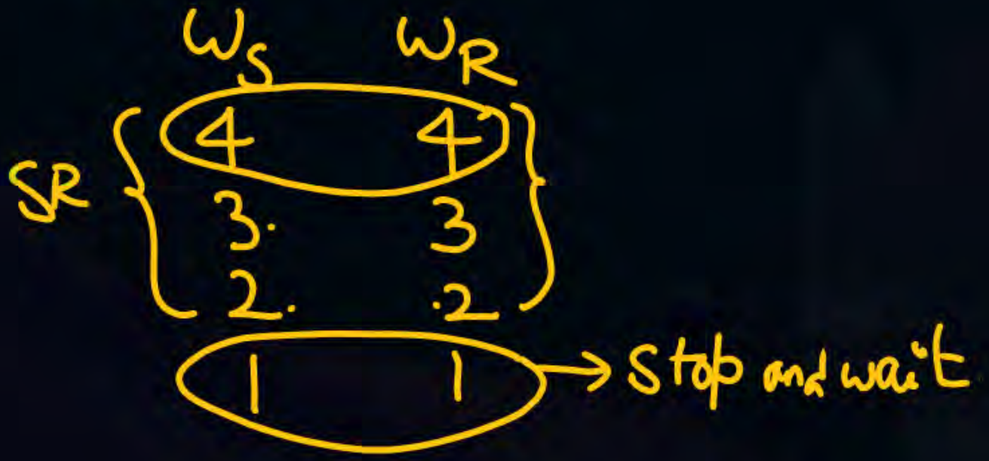

$W_S = W_R = 4$
3210
4567
$W_S + W_R \leq ASN$
$4 + 4 \leq ASN$
$ASN \geq 8$

For any SWP, always $|W_S| + |W_R| \leq ASN$

If $ASN = N$, then max W_S and W_R ? in SR

$$W_S = N/2 \qquad W_R = N/2$$

$$ASN = 8$$



If no of bits in seq number field is 'k'
then max W_S, max W_R in SR?

$\frac{2^k}{2}$

2^{k-1}

$W_S = 1$

2^{k-1}

$W_R = 1$

In SR $W_S = W_R = 4$, 10 packets are transmitted, then every 4th packet is lost, totally how many retransmissions?

1 2 3 4 4 5 6 7 7 8 9 10 10

(13) → (3)

Note:- SR uses negative acks



GBN



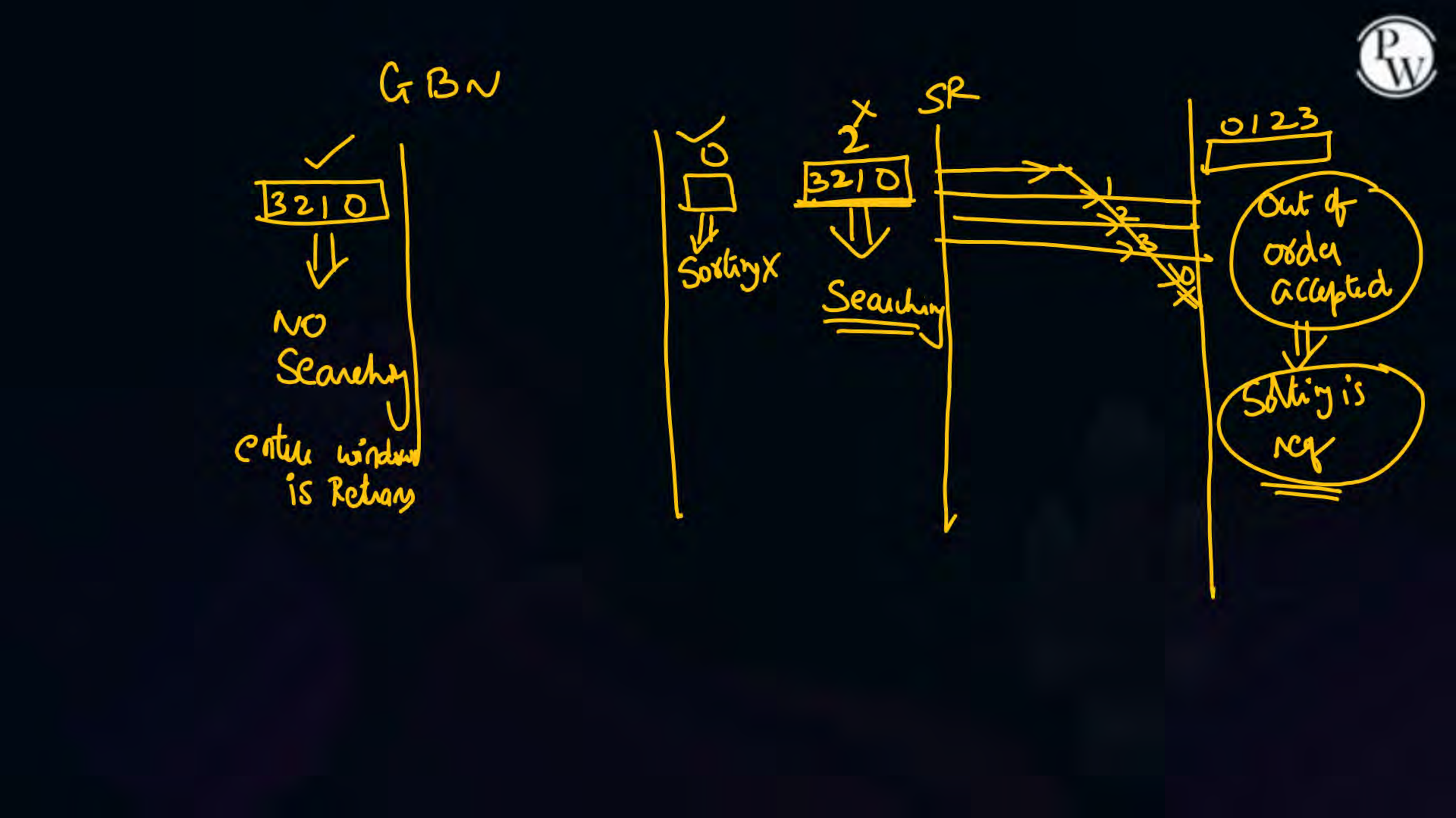
GBN Silently discards a corrupted packet

SR



SR will send a NAK. for corrupted packets

	Stop and wait	GBN	SR
Efficiency	$1/1+2a$	$N/1+2a$	$W_S/1+2a$
Buffers	$1+1$	$N+1$	$W_S+W_R = 2*W_S = 2*W_R$
min Seq number	2	$N+1$	W_S+W_R
Retransmissions	1	N	1
BW	low	high	low
CPU	low	moderate	high
implementation	very easy	moderate	complex


GBN
3210
NO
Searching
entire window
is Retran
0
Sorting x
2
3210
Searching
SR
0123
Out of
order
accepted
Sorting is
req

THANK - YOU